

7Aa DOCTORS PAST AND PRESENT

For thousands of years people have gone to see doctors when they feel unwell. Some of the ways in which doctors examine patients have not changed! For example, 3000 years ago, Ancient Egyptian doctors knew that if a person's **heart** was not beating as well as usual that person could be ill.

Today, doctors still find out how well your heart is beating. They may also measure temperature and do blood and urine tests to see if there are changes in your body compared to normal. These changes are called **symptoms**. Different problems cause different symptoms. The symptoms of a cold include a sore throat and runny nose.

B | Doctors have written instructions for how to treat problems for thousands of years. This Ancient Egyptian carving records the range of instruments and medicines that a doctor used to treat patients. Many of the instruments still look familiar!



A | Modern equipment can see **organs** inside our bodies. Scientists have used this technology on Egyptian mummies and found that many had heart problems.

A doctor sees if there is a match between a patient's symptoms and a known problem. If there is a match, the symptoms are **evidence** that the person has a certain illness.

Luckily, if you need a **medicine** today it will not contain a favourite ingredient of Ancient Egyptian medicine – animal dung!

- 1** a| A patient has a high temperature, a headache and a stuffy nose. Which word in bold on this page best describes these findings?
b| Suggest what illness the patient has.
- 2** A doctor tells a patient that they have acne. Suggest what evidence the doctor has found to make them think this.
- 3** a| Which of the following best describes the heart:
A| an organ B| a cell
C| a tissue D| a system?
b| State one job that the heart does.

LIFE 7Aa PROCESSES

WHAT DO ALL LIVING THINGS DO?

The Ancient Egyptians believed they had cures for death, including one made from onions. It is doubtful that this worked! When they died, the bodies of important people were treated to stop them rotting – they were mummified. This was done because Ancient Egyptians believed that living things contained a 'life force' called *ka*, which needed somewhere to live.

Today, we have different ideas about what it means to be alive. We look at what things do. If something can do the following **life processes**, it is a 'living thing' or **organism**:

- move
- grow
- need nutrition.
- reproduce
- respire
- sense things
- excrete waste



A | The mummies of some of the most important Ancient Egyptian rulers (the pharaohs) were placed inside huge stone pyramids.

1 Copy and complete the table below with the items in the list.

car, chair, coal, cow,
daffodil, goldfish,
mouse, octopus, robot,
rock, snake, Sun

Organism	Not an organism

2 A **mnemonic** is a word or phrase that helps you remember a list. It is usually made using the first letters of the words in a list. What mnemonic is spelled out by the first letters of the life processes?

Movement

All living things can either **move** from place to place or move parts of themselves.



B | Arctic poppies move parts of themselves. Their flowers turn to follow the Sun during the day.

Reproduction

Organisms can make more living things like themselves. We say that they can **reproduce**.



C | Offspring are the result of reproduction.

3 Suggest one difference between how most animals move and how most plants move.

4 Suggest one thing that many plants do to reproduce but animals do not do.

Sensitivity

Organisms **sense** and react to things around them.



D | This sensitive plant closes its leaves if it senses something touching them.

Growth

Living things increase in size. We say that they **grow**.



E | Some types of bamboo can grow 4 cm taller in an hour.

Respiration

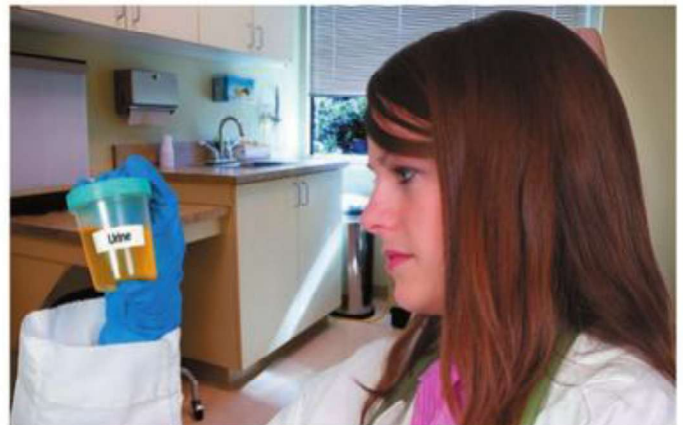
Living things use a process called **respiration** to release energy for them to use.



F | Humans, like many living things, need **oxygen** and food in order to respire.

Excretion

Organisms produce waste materials. When they get rid of these waste materials we say that they **excrete** them.



G | Waste materials include liquids (urine).

Nutrition

Living things require various substances to help carry out other life processes. We say that they need **nutrition**.

H | Animals eat food but plants make their own food. However, even plants need small amounts of substances from the soil to help them grow well.



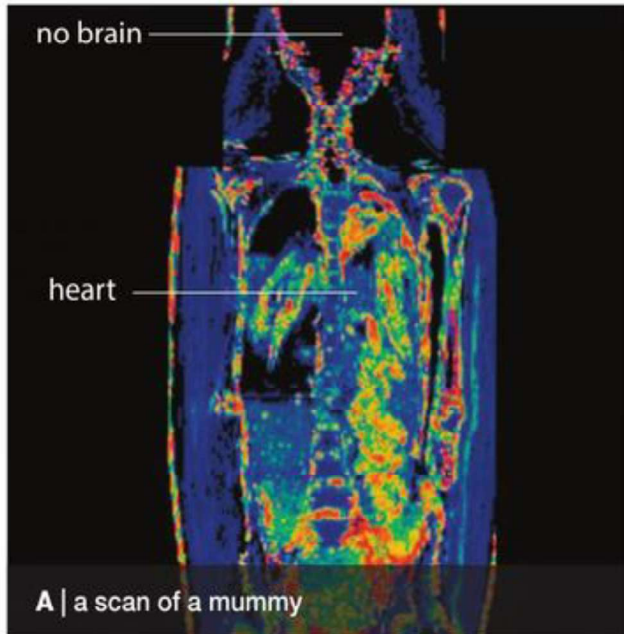
- 5** Describe two ways in which you show sensitivity.
- 6** Suggest one difference between how trees grow and how humans grow.
- 7** Suggest one difference between how fish and humans get their oxygen.
- 8**
 - a| In what ways is a car like an organism?
 - b| Why is a car not an organism?

I can ...

- recall and describe the life processes
- explain the differences between organisms and non-living things.

7Ab ORGANS

WHAT DO ORGANS DO?

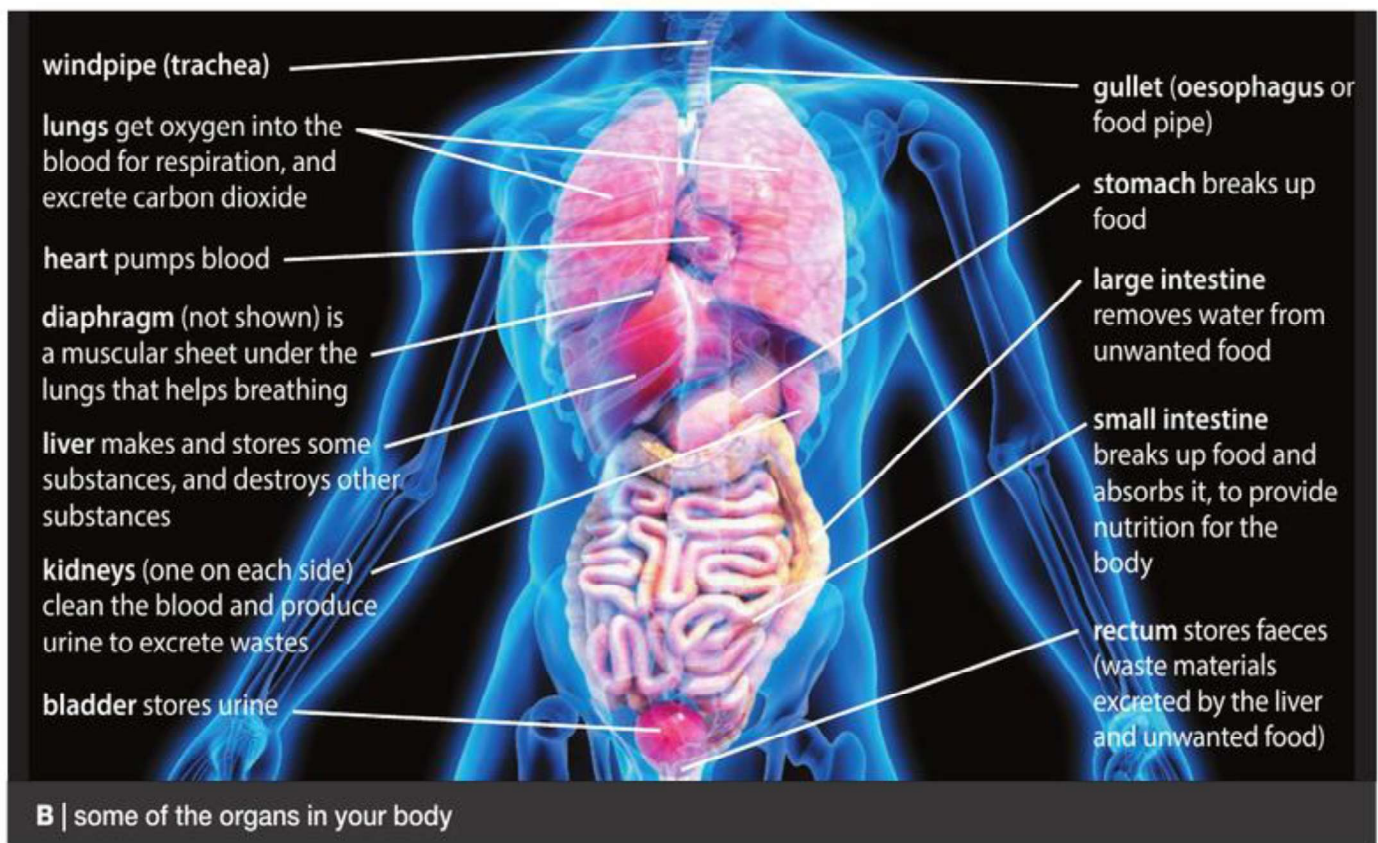


In Ancient Egypt, the heart was thought to be the most important part of a person. When people were mummified, the organs in their bodies were removed but the heart was left in place. The stomach, intestines, lungs and liver were thought to be useful on the person's journey in the afterlife and these organs were preserved. They did not think the **brain** was important and so it was often thrown away.

Human organs

The heart, stomach, intestines, lungs and liver are **organs**. Every organ has an important **function** (job). We now know that the brain is also an organ and has the very important function of controlling the body!

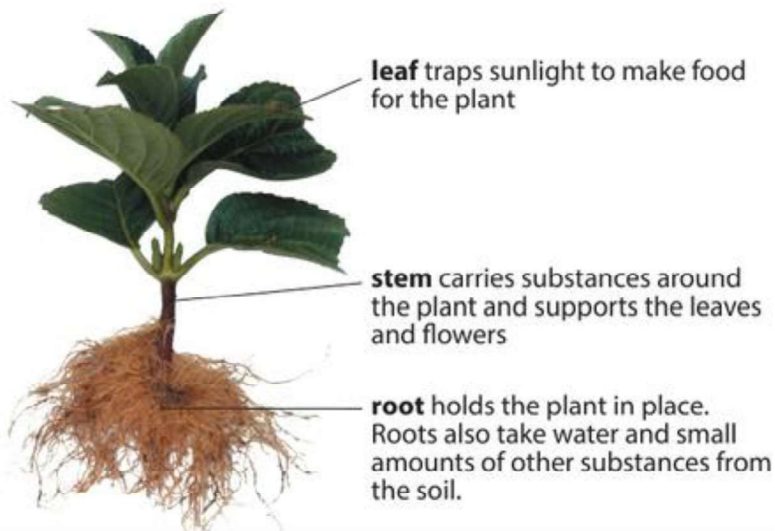
Your body's biggest organ is on the outside. It's your **skin**. Skin is used for protection and sensing things.



- 1 Draw a table to show the functions of five different organs in your body.
- 2 Which organ gets bigger as it fills with air?
- 3 List the organs that help to get nutrition into the body.
- 4 List the organs that excrete waste materials.
- 5 List two organs that store solid or liquid wastes.

Plant organs

Photo D shows some of the main organs in plants.



D | the organs found in most plants

Plants make their own food using a process called **photosynthesis**. This process occurs in the leaves when there is light. Photosynthesis needs **carbon dioxide** from the air, and water. Some plants also have **storage organs**, which they use to store some of the food that they make. Potatoes and carrots are storage organs.

- 6 Which organ is the main organ of nutrition in a plant?
- 7 a| What process produces the food stored in plant storage organs?
b| Why won't a potato grow if the potato plant does not get much light?
- 8 Which human organ is most similar to a leaf? Explain your reasoning.

FACT



Leaves are plant organs that are designed to collect sunlight. Plants that live in shady areas often have very large leaves. The leaves of the giant water lily can be up to 3 m in diameter.



E | Many plants have underground storage organs, which they use to store some of the food that they make. Humans often eat these.

I can ...

- identify and locate important plant and animal organs
- describe the functions of important plant and animal organs
- describe what happens in photosynthesis.

MEDICAL 7Ab DOCTORS

HOW DO MEDICAL DOCTORS USE STEM SKILLS?

STEM stands for science, technology, engineering and maths. There are many interesting jobs open to people with skills in these subjects. These include careers in communications, farming, fashion, films, finance, health, sport and video gaming.

All STEM subjects are linked:

- similar skills are needed for each one
- changes or advances in one subject may change the way the others are done.

STEM skills

An important STEM skill is problem-solving. This is often done by thinking up ideas and then testing them. Results from the tests are used as **evidence** (information used to decide if an idea is correct or incorrect). A decision made using evidence is called a **conclusion**.

Doctors think up ideas about what might be wrong with a patient. They then perform tests. They use the test results and their knowledge of the human body to make a conclusion (called a 'medical diagnosis').

- 1** List the STEM subjects.



B | a doctor performing a test



A | X-ray technology was invented in 1895 and allowed doctors to see the bones inside people. Advances in the technology have produced scanners (like this one) that show the organs inside our bodies. Engineers also use X-ray technology to check joins between pieces of metal.

- 2** a | When a girl broke her leg, it was put in plaster for 2 months. Suggest how a doctor might check to see if the two ends of the broken bone have now joined together?
b | A doctor thinks that a patient has a heart problem. Suggest why the patient is put in an X-ray scanner.
- 3** Look at photo B. What organ is the doctor testing?
- 4** A man has been getting pains in his chest. His doctor thinks the man may have a heart problem.
a | What does the heart do?
b | Suggest two tests the doctor uses to discover if the man's heart is working properly.
- 5** A patient has yellow skin. The doctor thinks the patient's liver has a problem and is releasing a substance called bilirubin into the blood. A test finds bilirubin in the patient's urine. The doctor says that the patient has liver disease.
a | What does the liver do?
b | What is the doctor's diagnosis?
c | What evidence does the doctor use?
d | Which organ produces urine?
e | Which organ stores urine?

STEM careers

Many other interesting jobs need a knowledge of the human body and its organs. Pathologists use their knowledge of the body and problem-solving skills to reach conclusions about how someone has died.



C | physiotherapists at work

Physiotherapists help people to recover from muscle and bone injuries. Physiotherapists are an important part of your country's sporting teams.

6 Explain why a country's athletics team includes many physiotherapists.

To develop their skills and knowledge, people who want to become physiotherapists, nurses or doctors do more training after they leave school. Doctors, for example, usually train at a university for six or seven years. Some doctors then do even more training to specialise in a certain area of the body or to become surgeons or pathologists.

7 Find out what a cardiologist is.

ACTIVITY

Work with others in your group to think about the different jobs that need a knowledge of the human body and its organs. Write down as many jobs as you can think of. For each job, give a reason why a knowledge of the human body is useful.

PRACTICAL

A doctor uses a stethoscope to listen to a person's heart beating and their breathing. The first stethoscope was a rolled-up tube of paper. In your group, make a range of different stethoscopes using paper, different types of tubing, funnels and sticky tape. Test them out and decide which one works best.



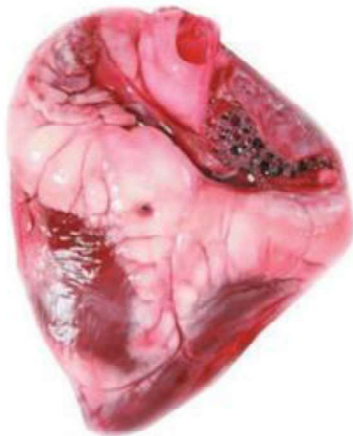
D | a modern stethoscope

7Ac TISSUES

WHY ARE TISSUES IMPORTANT?

Many good detective stories have a 'pathologist', who inspects a dead body to look for evidence to help to solve a murder. Pathologists have a long history; dead bodies were examined in Ancient Egypt, Ancient Greece and in Roman times.

Pathologists are fully trained doctors. Some pathologists examine dead bodies to try to work out causes of death. Others examine small pieces taken from living people to try to identify diseases.



B | a human heart

A pathologist will look at a heart in detail because it is such an important organ and damage to it often causes death. Its function is to pump blood around the body. The blood carries oxygen and nutrients (from food) for all the different parts of the body to use.

The heart has different parts. In photo B, the whiter parts are fat and the reddish parts are muscle. These are **issues**. All organs are made up of different tissues.

Each tissue in an organ has a certain function. For example, the **muscle tissue** in the heart is the part that moves, to pump blood. The **fat tissue** helps to protect the heart.

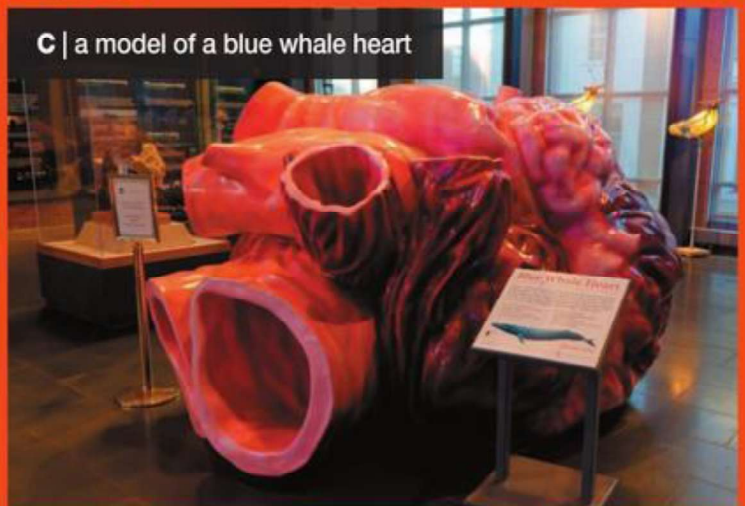


A | Pathologists use microscopes, which magnify things. This allows pieces taken from a body to be examined in great detail.

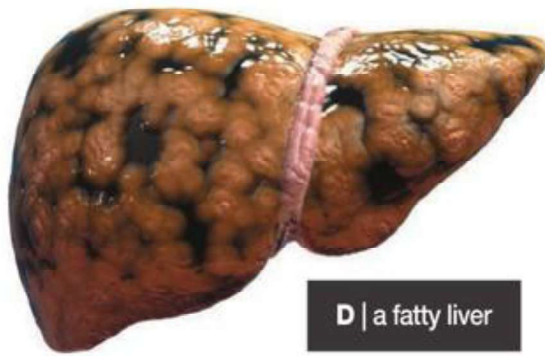
- 1** What does a microscope do?
- 2** Look at photo B. Describe what a heart looks like.

FACT

C | a model of a blue whale heart



About 20 per cent of the mass of a mammal heart is fat. For an adult human, that is about 60 g of fat. For an adult blue whale that is about 120 kg of fat; a blue whale heart can have a mass up to about 600 kg!

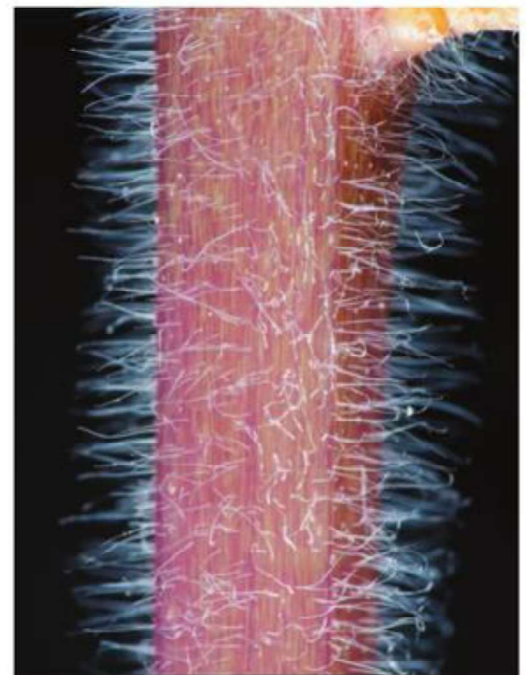


- 3 a | Name two tissues found in the heart.
b | What does each of these tissues do?
- 4 A pathologist says that the liver in photo D comes from someone with 'fatty liver disease'. Compare it with the healthy liver in diagram B on page 8. What evidence supports the pathologist's conclusion?
- 5 Your intestines move, so that food is pushed along. What tissue would you expect to find in intestines?

Plant tissues

Plants also have organs made out of tissues. Many roots, like the one shown in photo E, have hairs on the outside. This is **root hair tissue** and it helps the root to take water out of the soil quickly.

If you cut open a plant organ, you can see more tissues. Photo F shows that a carrot contains different tissues. The tissue in the middle of the carrot is called **xylem tissue** (pronounced 'zy-lem'). Xylem tissue carries water. In a carrot, the xylem tissue carries water up from the roots, through the carrot and on into the rest of the plant.



E | root hair tissue on a radish plant

xylem tissue



F | Carrots contain different tissues.



G | a carrot plant

- 6 a | What sort of organ is a carrot?
b | How many tissues does a carrot contain? Explain your reasoning.
- 7 a | Name two tissues you would expect to find in a radish plant root.
b | What does each of these tissues do?
- 8 Name a plant organ that is above ground and contains xylem tissue.
- 9 Which life process does xylem tissue help with?

I can ...

- identify and recall named tissues in human and plant organs
- describe the functions of different tissues in an organ.

7Ac MICROSCOPES

HOW IS A LIGHT MICROSCOPE USED TO EXAMINE A SPECIMEN?

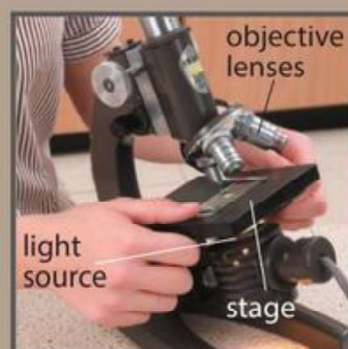
To find out what is wrong with an organ, doctors do tests. Some tests involve taking a small piece of tissue (a biopsy) from an organ and looking at it under a light microscope. Microscopes make things appear bigger; they magnify things. The Method below shows how to use a light microscope.

Method

- A** | Place the smallest **objective lens** (the lowest **magnification**) over the hole in the stage. Turn the **coarse focusing wheel** to make the gap between the objective lens and the stage as small as possible.



- B** | Place the **slide** under the clips on the **stage**. The slide contains the **specimen** (the thing you want to look at). Then adjust the light source so that light goes up through the hole.



- C** | Look through the **eyepiece lens**. Turn the coarse focusing wheel slowly until what you see is in focus (clear and sharp).



- D** | To see a bigger **image**, place the next largest objective lens over your specimen.



- E** | Use the **fine focusing wheel** to get your image in focus again. Do not use the coarse focusing wheel since you can break the slide and damage the objective lens. If you cannot see your specimen clearly go back to a lower magnification.



Never point a microscope mirror at the Sun. This can permanently damage your eyesight.

- 1** How many types of lenses are found in a light microscope?
- 2** Write down some rules of your own for:
 - a| using a microscope safely
 - b| taking care of a microscope.
- 3** What part of a microscope makes the image clearer?
- 4** What is a specimen?

WORKING SCIENTIFICALLY

Both of the lenses in a light microscope do some magnifying. How much a lens magnifies is written on its side (e.g. $\times 10$). To work out the total magnification of both lenses working together, we use this formula:

total magnification = magnification of eyepiece lens $\times$ magnification of objective lens

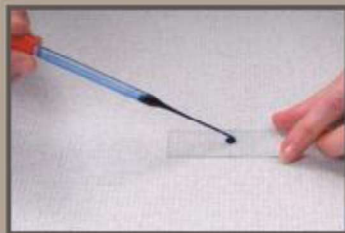
- 5** A microscope has a $\times 10$ eyepiece lens and a $\times 15$ objective lens. What is its total magnification?

Preparing a specimen

The specimen on a microscope slide needs to be thin so that light can pass through it. A thin, glass **coverslip** is put on the specimen to keep it flat, hold it in place and stop it drying out. The Method below shows how to prepare a slide of onion tissue.

Method

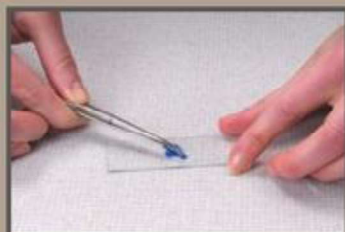
- A** | Take a slide and place a drop of water in the centre. The water may contain a **stain** to make the specimen show up better.



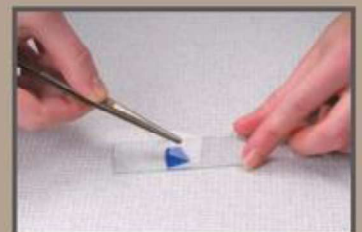
- B** | Use some forceps to peel off the inside layer of a piece of onion.



- C** | Place a small piece of onion skin onto the drop of water on your slide.



- D** | Use some forceps to lower a coverslip onto your specimen. If you do this carefully and slowly you will not get air bubbles trapped under the coverslip.



- 6** Why does a specimen need to be thin?
- 7** Why do we use coverslips?
- 8** Suggest the names of two plant and two animal tissues you could examine using a light microscope.
- 9** Plan an investigation to examine rhubarb stem tissue in detail.
- 10** Jake sets up a microscope but only sees darkness when looking into the eyepiece lens. What might be wrong? Write down as many things as you can think of.



Wear eye protection when carrying out this method. Slides and coverslips are made of thin glass. Be very careful when using them.

I can ...

- describe how to prepare a microscope slide
- describe how to use a light microscope to examine a specimen.